In [3]:
```
import pandas as pd
import numpy as np

np.random.seed(42)

db = pd.DataFrame({
    'Pregnancies': np.random.randint(0, 10, 100),
    'Glucose': np.random.randint(70, 200, 100),
    'BloodPressure': np.random.randint(50, 100, 100),
    'SkinThickness': np.random.randint(10, 50, 100),
    'Insulin': np.random.randint(0, 300, 100),
    'BMI': np.random.uniform(18, 45, 100).round(1),
    'DiabetesPedigreeFunction': np.random.uniform(0.1, 2.5, 100).round(3),
    'Age': np.random.randint(21, 80, 100),
    'Outcome': np.random.randint(0, 2, 100)
})

print(db.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6       81             97             12       22  27.5
1            3      102             61             49      249  21.2
2            7      117             54             33        9  21.9
3            4      131             86             41      245  38.6
4            6      106             81             31       99  34.7

   DiabetesPedigreeFunction  Age  Outcome
0                     0.235   34        0
1                     2.175   79        1
2                     2.051   76        1
3                     2.499   71        1
4                     2.492   27        0
```

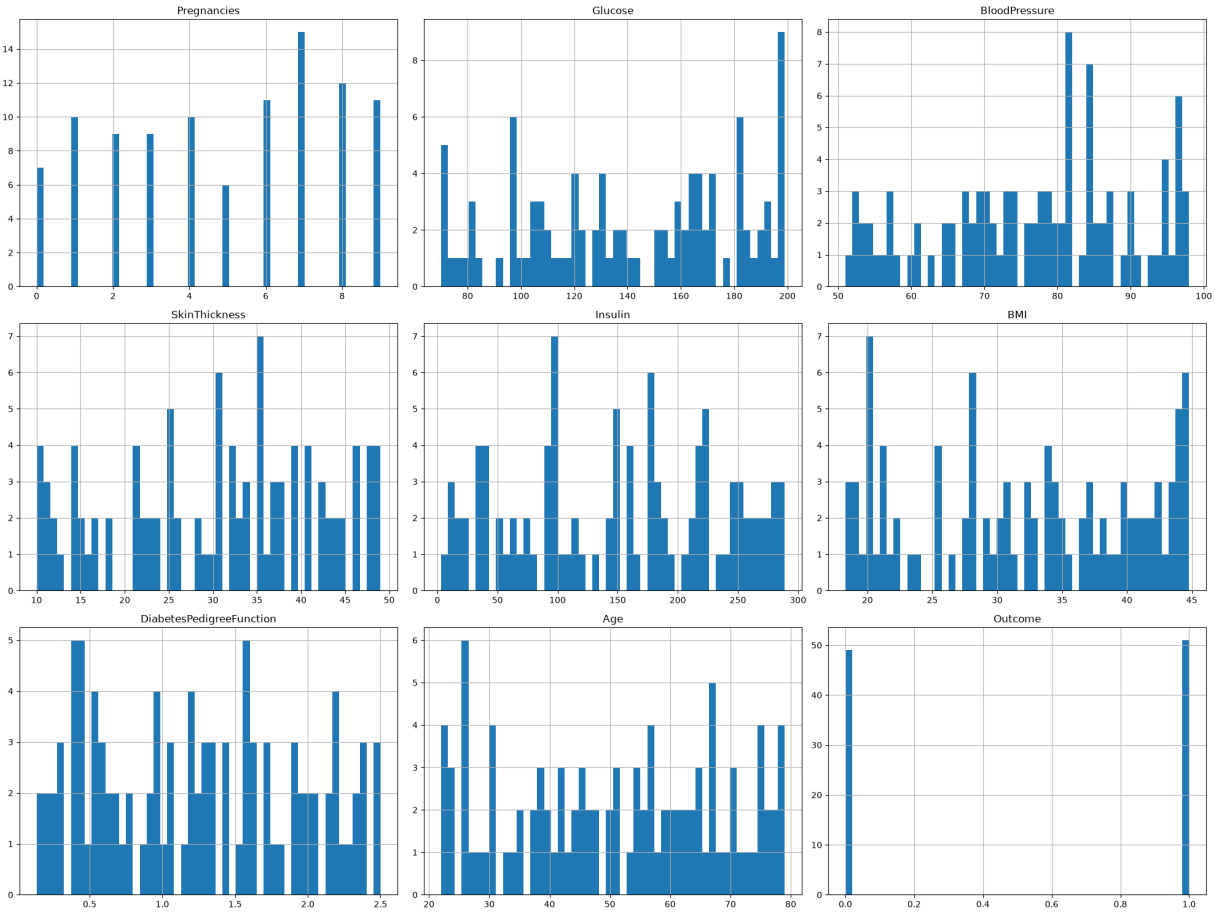
In [4]:
```
print(db.info())
print(db.describe())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               100 non-null    int32  
 1   Glucose                   100 non-null    int32  
 2   BloodPressure             100 non-null    int32  
 3   SkinThickness             100 non-null    int32  
 4   Insulin                   100 non-null    int32  
 5   BMI                       100 non-null    float64
 6   DiabetesPedigreeFunction  100 non-null    float64
 7   Age                       100 non-null    int32  
 8   Outcome                   100 non-null    int32  
dtypes: float64(2), int32(7)
memory usage: 4.4 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   100.000000  100.000000     100.000000     100.000000  100.000000   
mean      4.910000  141.520000      76.950000      30.660000  151.780000   
std       2.871296   39.943769      13.391822      11.669887   84.769036   
min       0.000000   70.000000      51.000000      10.000000    3.000000   
25%       2.000000  107.750000      67.750000      21.750000   87.500000   
50%       5.000000  142.000000      79.000000      32.000000  160.500000   
75%       7.000000  174.000000      86.000000      39.500000  222.000000   
max       9.000000  199.000000      98.000000      49.000000  289.000000   

              BMI  DiabetesPedigreeFunction         Age     Outcome  
count  100.000000                100.000000  100.000000  100.000000  
mean    32.384000                  1.243570   51.180000    0.510000  
std      8.613892                  0.696678   17.420167    0.502418  
min     18.300000                  0.135000   22.000000    0.000000  
25%     25.475000                  0.586500   37.000000    0.000000  
50%     33.050000                  1.224000   52.000000    1.000000  
75%     40.200000                  1.816000   66.250000    1.000000  
max     44.800000                  2.499000   79.000000    1.000000  
```

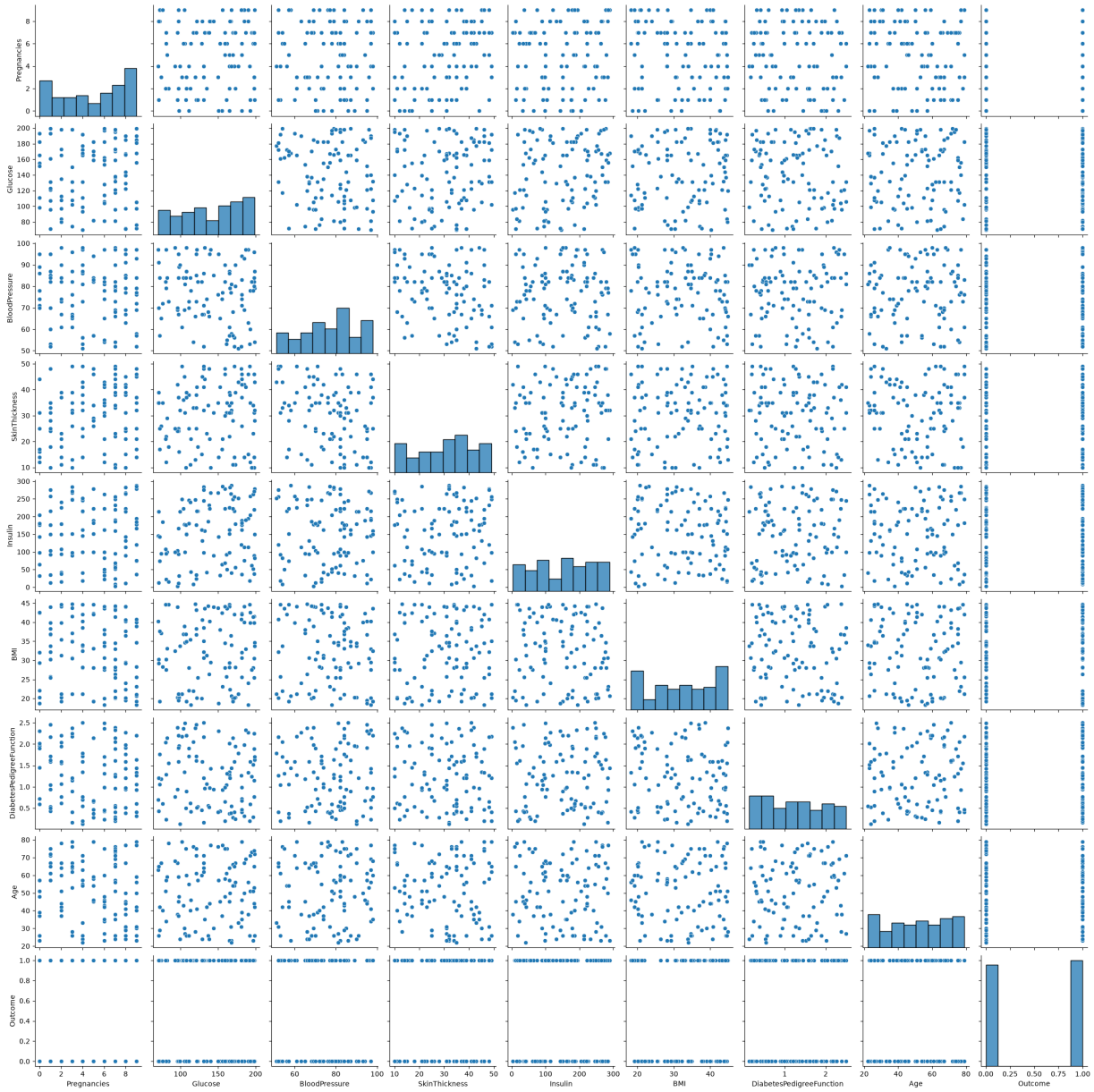
In [5]:
```
import matplotlib.pyplot as plt

db.hist(bins=50, figsize=(20, 15))
plt.tight_layout()
plt.show()
```



In [6]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(db)
plt.show()
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

In [ ]: